Mexico's Health Transition: Life Expectancy, Causes of Death and Risk Factors in WHO Data

Executive EDA Report

Projecto for time series, Units 1 & 2

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Infographic

EXECUTIVE EDA REPORT · WHO DATA FOR MEXICO

Mexico's Health Transition

Life expectancy, causes of death and risk factors — what 24 years of WHO indicators (and 74 years of population data) show

LIFE EXPECTANCY AT BIRTH

70.8 years (2021)

Rose from 74.2 to 75.9 (2000–2019), then fell **−4.9 years in 2020** with the pandemic.



POPULATION GROWTH RATE

0.9% per year (2023)

Down from a peak of 3.3% in 1966. Population reached **129.7 million** in 2023.



LEADING CAUSE OF DEATH

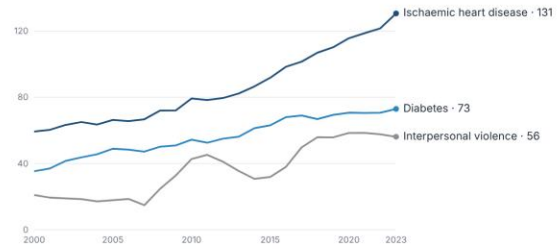
131 per 100,000 (2023)

Ischaemic heart disease: #1 every year except 2020–21. Rate **+120%** since 2000.



Mortality has shifted toward chronic disease

Death rates per 100,000 people for the causes with the largest changes, 2000–2023.



≈345

COVID-19 deaths per 100,000 in both 2020 and 2021 — the #1 cause for two years, down to 70 in 2022 and out of the top 10 in 2023.

2×

Diabetes deaths doubled: 35 → 73 per 100,000

4×

Violence deaths: 15 (2007) → 58 (2021)

Risk factors: first year → latest year

WHO modelled estimates for the total population (both sexes).

Adult obesity (18+)	**Alcohol per capita (15+)**	**Safe drinking water**	**Hypertension (30–79)**	**Tobacco use (15+)**
2000 → 2022	2000 → 2022	2000 → 2024	2000 → 2019	2000 → 2024
21.6% ▲ **36.0%**	5.1 L ▲ **6.0 L**	40.0% ▲ **43.0%**	34.0% ▼ **32.1%**	21.2% ▼ **15.7%**
+14.4 pts	+0.9 L	57% still without	peak 35.1% in 2008–09	−5.5 pts

What the data show

Trend dominates

Every series is driven by a long-term trend. Each year looks very much like the previous one.

No seasonality or cycles

Annual data cannot show seasonal patterns, and no regular multi-year cycle appears.

One major shock

COVID-19 (2020–2021) is the main outlier: it broke the life expectancy trend but not the rise of chronic disease.

Sources: World Health Organization, data.who.int (life expectancy, causes of death — Global Health Estimates, risk factors); UN World Population Prospects (population). Latest year available differs by indicator (2019–2024). Analysis: exploratory only, no forecasting.

Executive summary

Mexico is living through a health transition: people live longer than a generation ago, but they increasingly die from chronic, lifestyle-related diseases. This report explores official World Health Organization (WHO) data to show how that transition looks in numbers, using exploratory data analysis (EDA) only, with no forecasting.

What data were analysed. Three groups of annual indicators for Mexico, all for the total population (both sexes):

Group	Indicator	Unit	Years	Source
Population	Population growth rate	% per year	1950–2023	UN World Population Prospects (used by WHO)
Population	Life expectancy at birth	years	2000–2021	WHO, indicator WHOSIS_000001
Mortality	Top 10 causes of death	deaths per 100,000	2000–2023	WHO Global Health Estimates (yearly ranking built for this report)
Risk factors	Hypertension, adults 30–79	% of adults	1990–2019	WHO (NCD_HYP_PREVALENCE_A)
Risk factors	Tobacco use, age 15+	% of population	2000–2024 (+ projections)	WHO (M_Est_tob_curr_std)
Risk factors	Alcohol consumption, age 15+	litres of pure alcohol per capita	2000–2022	WHO (SA_0000001688)
Risk factors	Obesity, adults 18+	% of adults	1990–2022	WHO (NCD_BMI_30A)
Risk factors	Safely managed drinking water	% of population	2000–2024	WHO/UNICEF JMP (WSH_WATER_SAFELY_MANAGED)

All data were downloaded from the WHO country page for Mexico (data.who.int/countries/484). Every series has one value per year, so each lag in the analysis equals one year.

Main findings.

- **Long-term trends dominate every series.** Population growth fell from 3.3% (1966) to 0.9% (2023), and life expectancy rose slowly from 74.2 to 75.9 years (2000–2019) before stalling.
- **COVID-19 was the largest shock in the data.** Life expectancy dropped 4.9 years in 2020, to 70.8 years in 2021, and COVID-19 was the leading cause of death in 2020 and 2021 (about 345 deaths per 100,000).
- **Mortality has shifted toward chronic disease.** Deaths from ischaemic heart disease (+120%) and diabetes (+107%) more than doubled between 2000 and 2023, and both kept rising during the pandemic.
- **Risk factors move in different directions.** Adult obesity rose from 21.6% to 36.0% and alcohol consumption increased, while tobacco use and, since 2009, hypertension declined. Only 43% of the population uses safely managed drinking water.

- **No seasonality and no regular cycles.** Annual data cannot show seasonal patterns, and no series repeats a pattern at a fixed interval.

Why it matters for decision-making. The data point to where health spending and prevention will be needed: chronic diseases tied to obesity and diet (heart disease, diabetes, kidney disease), violence prevention, and basic services such as safe water. They also show a practical limit: the most recent WHO figures for life expectancy end in 2021, so current decisions rely on data that are several years old.

Exploratory data analysis

Each series was examined for the five features the analysis requires: trend, seasonality, cyclic behaviour, outliers and changes of pattern over time. For each one the report shows the series over time with its trend (a centred moving average), the year-over-year change and the remainder (what is left after removing the trend), followed by lag plots and the autocorrelation function (ACF). Because the data are annual, there is no seasonal component to separate: seasonality is a pattern that repeats within a year (months or quarters), which one value per year cannot show.

1. Population growth rate (1950–2023)

Mexico's population growth rate follows one clear, long-term trend: it rose to a peak of 3.3% per year in 1966 and has been falling ever since, reaching 0.9% in 2023. Over the same period the population grew from 27.6 to 129.7 million people, so the country is still growing, but much more slowly.

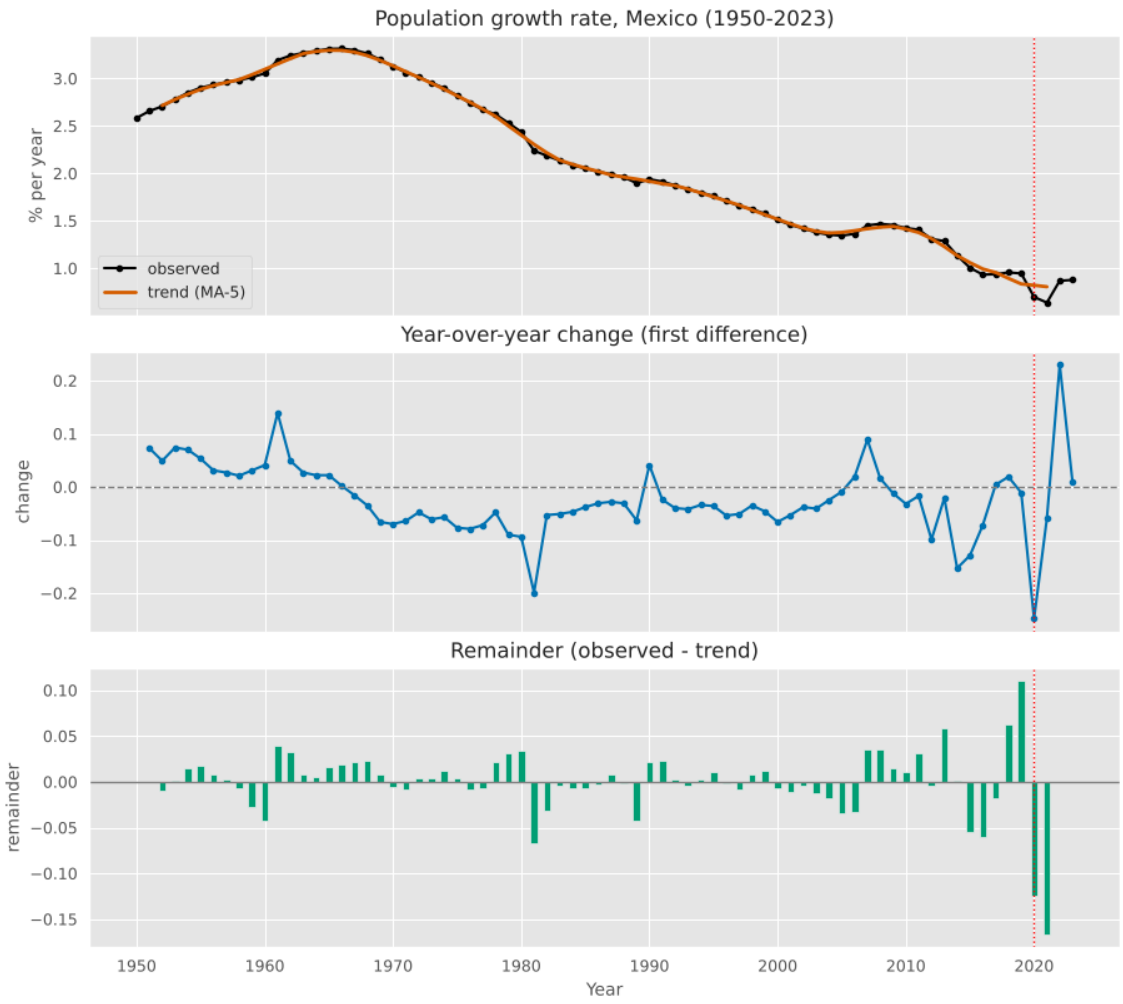


Figure 1. Population growth rate with its 5-year moving-average trend (top), year-over-year change (middle) and remainder (bottom).

- **Trend:** strong and smooth. The decline is steady from the late 1960s to the early 2000s, pauses around 2005–2009 and speeds up again after 2012.
- **Seasonality:** not applicable with annual data.
- **Cycles:** none. The pause of 2005–2009 and the faster decline of 2012–2016 are single episodes, not a pattern that repeats.
- **Outliers:** 2020 and 2021 are the lowest values of the whole series (0.70% and 0.64%) and the largest remainders, followed by a rebound to 0.87% in 2022. This matches the excess deaths of the pandemic. A smaller one-year drop appears in 1981.

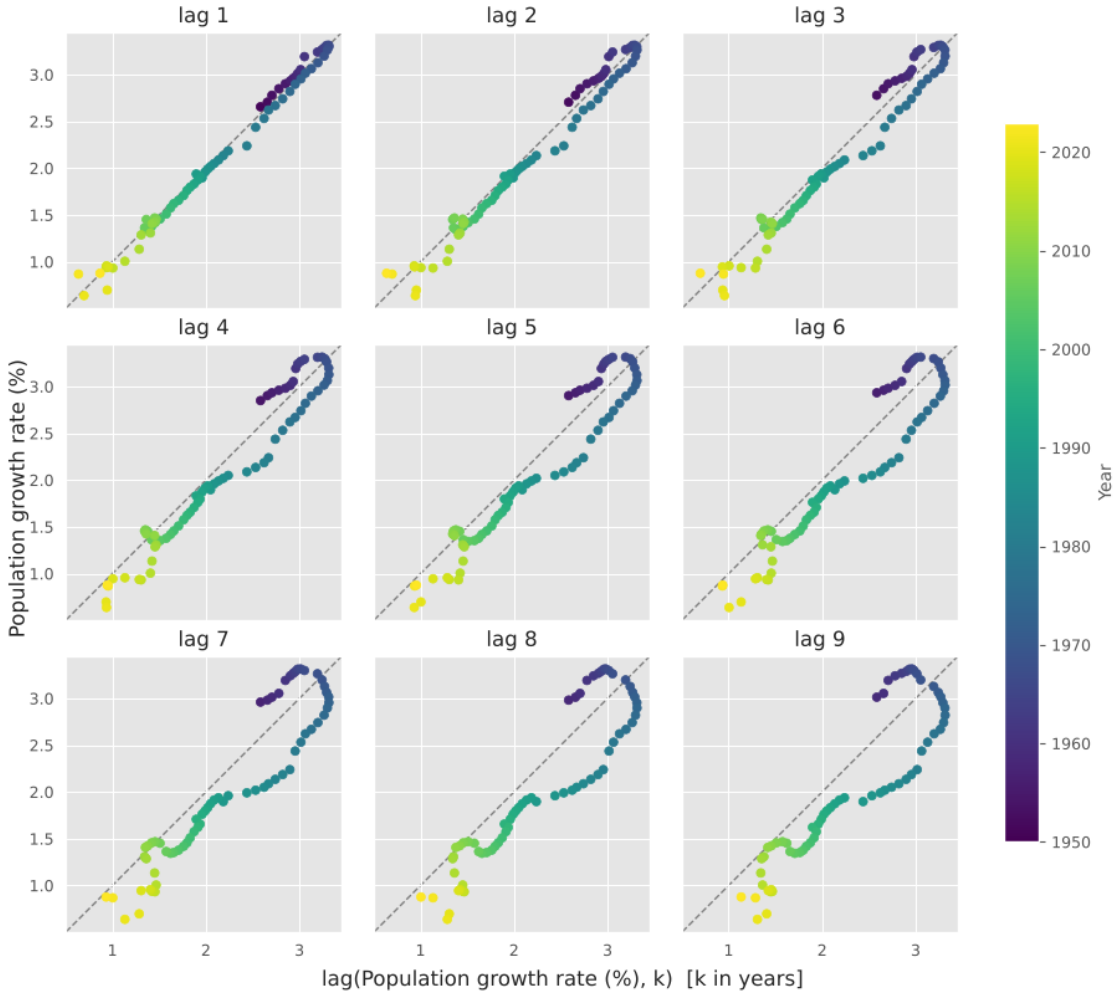


Figure 2. Lag plots: each point compares one year's value (vertical axis) with the value k years earlier (horizontal axis). Colour shows the year.

At lag 1 the points sit almost exactly on the diagonal: each year is very similar to the year before, which means the series has strong memory (positive autocorrelation). As the lag grows, the points form a loop. The early years (purple, 1950s–1960s) sit above the diagonal because growth was still increasing; from the 1970s onward the points fall below it because growth was lower than k years before. The loop is the turning point of the trend in 1966, not a seasonal or cyclic pattern.

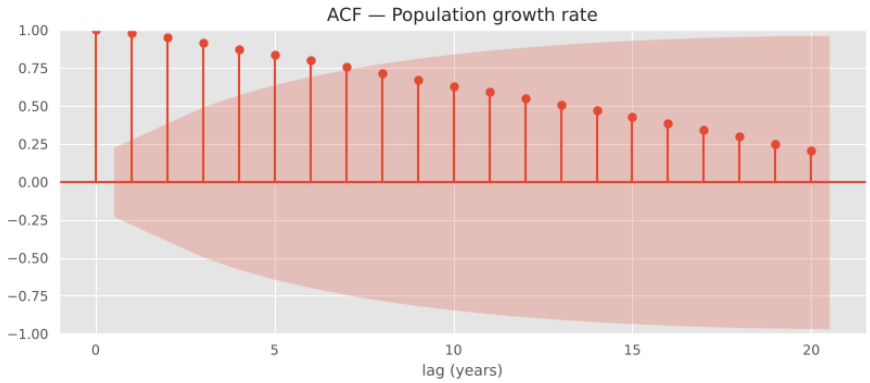


Figure 3. Autocorrelation function (ACF). The shaded band marks values that are not statistically different from zero.

The ACF starts close to 1 and decreases slowly, staying positive for 20 lags. This slow decay is the typical signature of a series driven by a trend, so the series is not stationary: its average level changes over time.

2. Life expectancy at birth (2000–2021)

Life expectancy in Mexico grew slowly for two decades and then collapsed with the pandemic: from 74.2 years in 2000 it reached 75.9 years in 2013, stayed flat until 2019 (75.8), and fell to 70.9 in 2020 and 70.8 in 2021. The WHO series ends in 2021, so the data do not yet show whether it recovered.

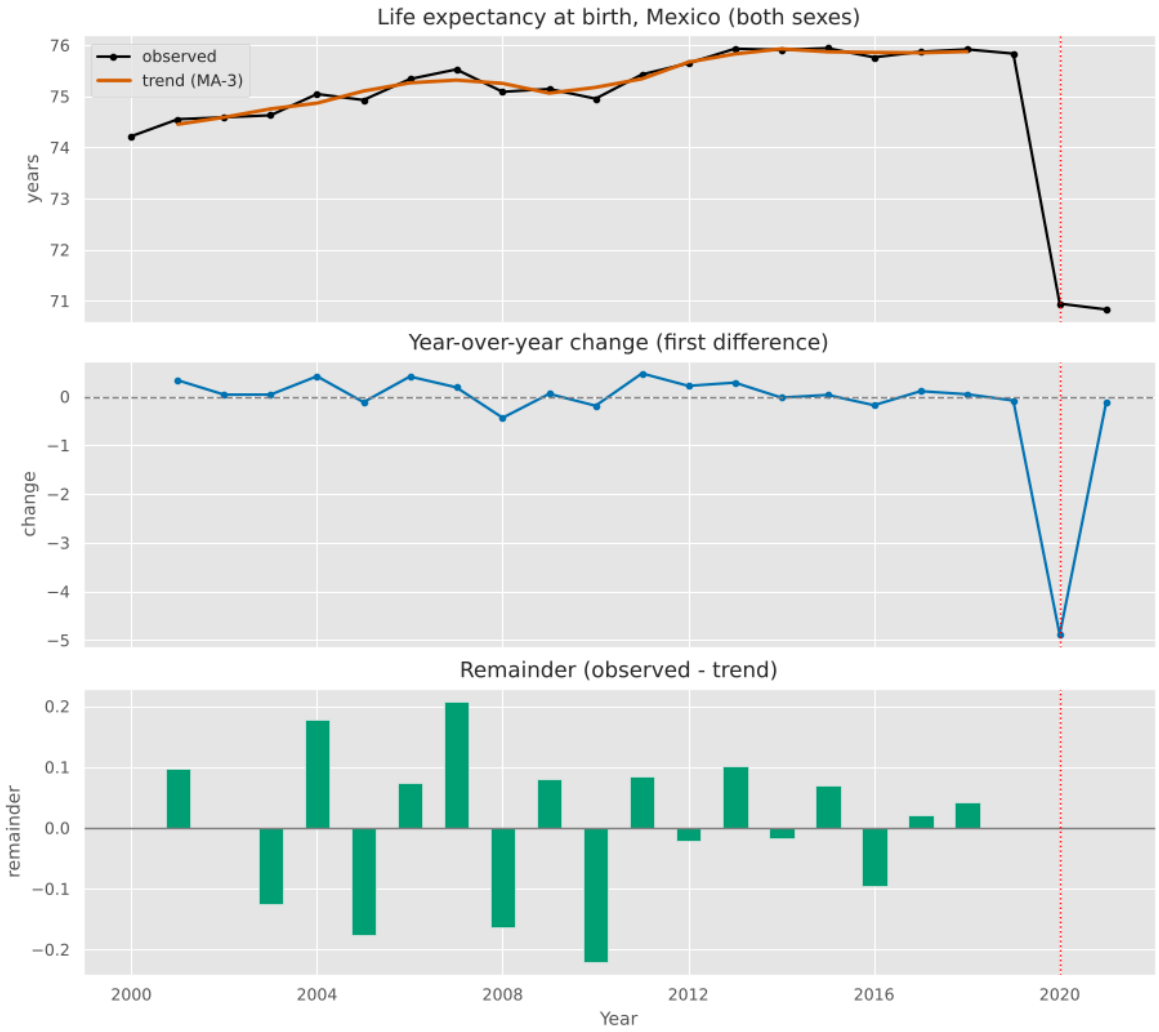


Figure 4. Life expectancy with its 3-year trend (computed separately before and after 2020, so the shock is not spread over neighbouring years), year-over-year change and remainder.

- **Trend:** upward but slow, about +0.1 years per year between 2000 and 2013, followed by a plateau from 2013 to 2019. The gains of the early 2000s had already stopped before the pandemic.
- **Change of pattern:** a dip between 2008 and 2010 (from 75.5 to 75.0 years). It coincides with the sharp rise in deaths from interpersonal violence in the same years (Section 3).
- **Outlier / structural break:** 2020, with a fall of −4.9 years in a single year. In 2021 the value stays at that low level, so it is a shift that lasted two years, not a one-off spike.

- **Seasonality and cycles:** none. Outside the shock, the remainder is only about ±0.2 years and alternates without a pattern, which is ordinary year-to-year noise.

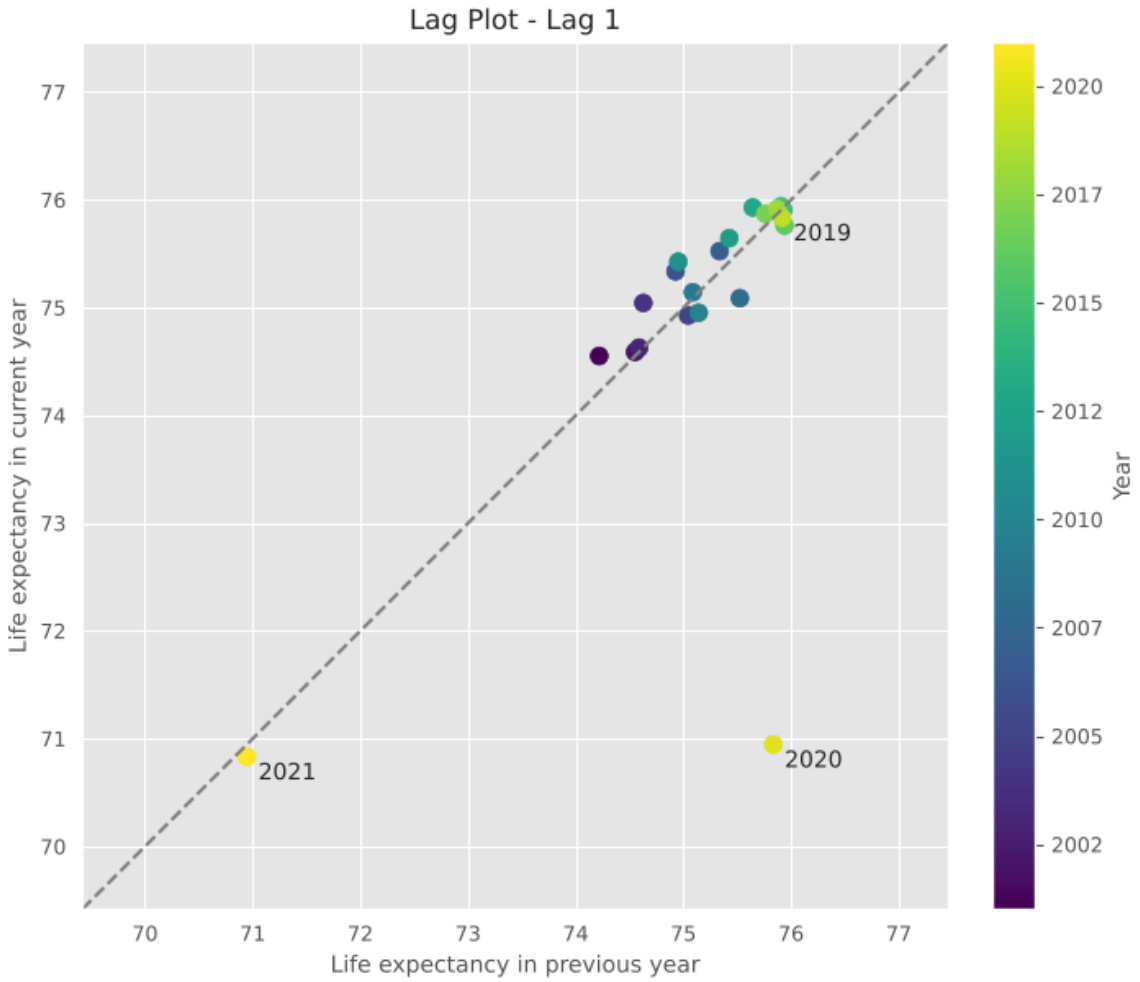


Figure 5. Lag-1 plot: life expectancy in each year against the previous year.

All years from 2000 to 2019 lie close to the diagonal: life expectancy in one year is a very good guide to the next. The 2020 point falls far below the diagonal, because the value dropped sharply from the year before. The 2021 point returns to the diagonal, but at a much lower level: the pandemic did not create one bad year, it moved the whole series down.

The 2020 drop is so large that it compresses the full lag plots and hides the normal behaviour of the series. For that reason the lag plots and the ACF were repeated for the pre-pandemic period, 2000–2019.

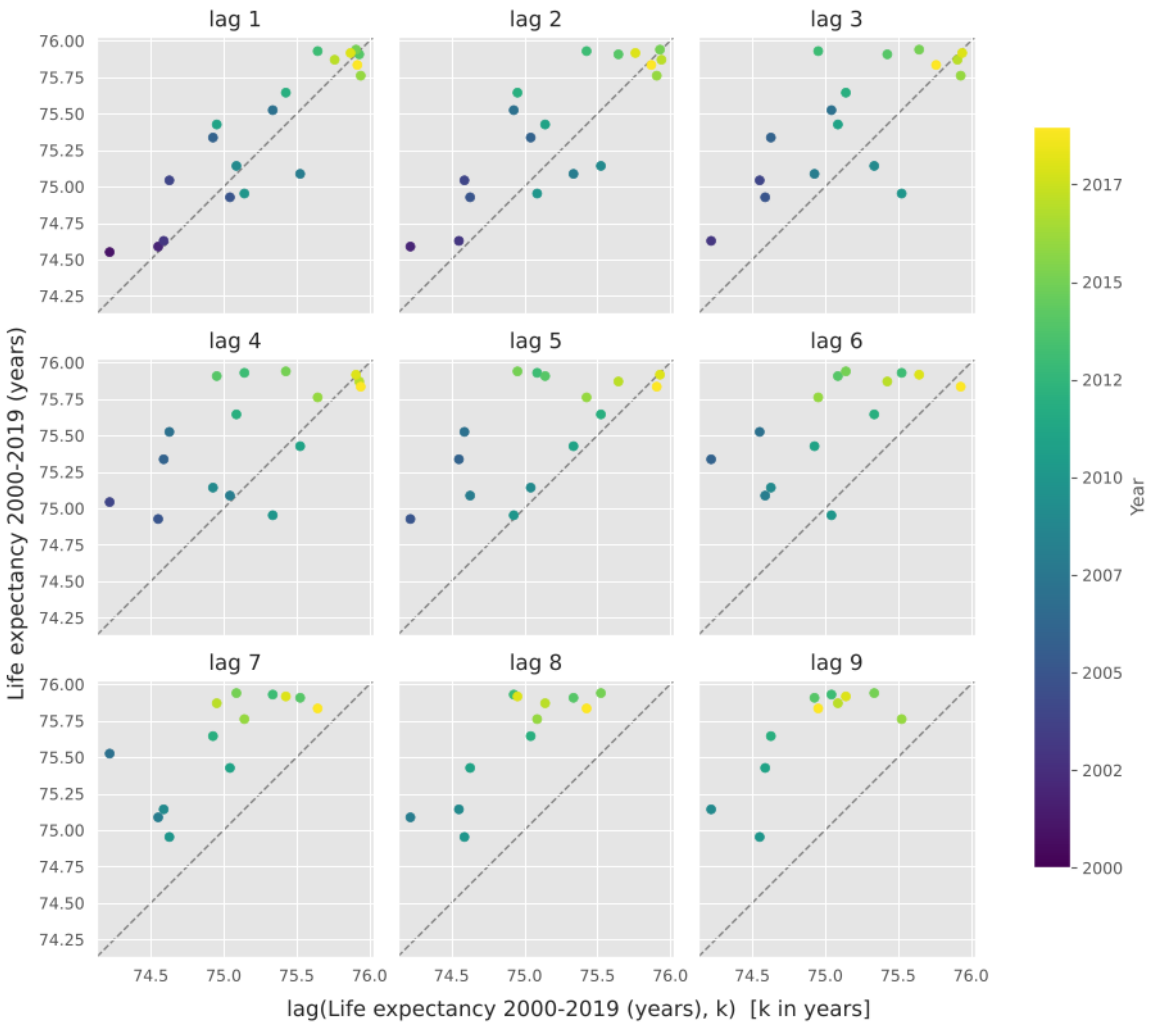


Figure 6. Lag plots of life expectancy, 2000–2019.

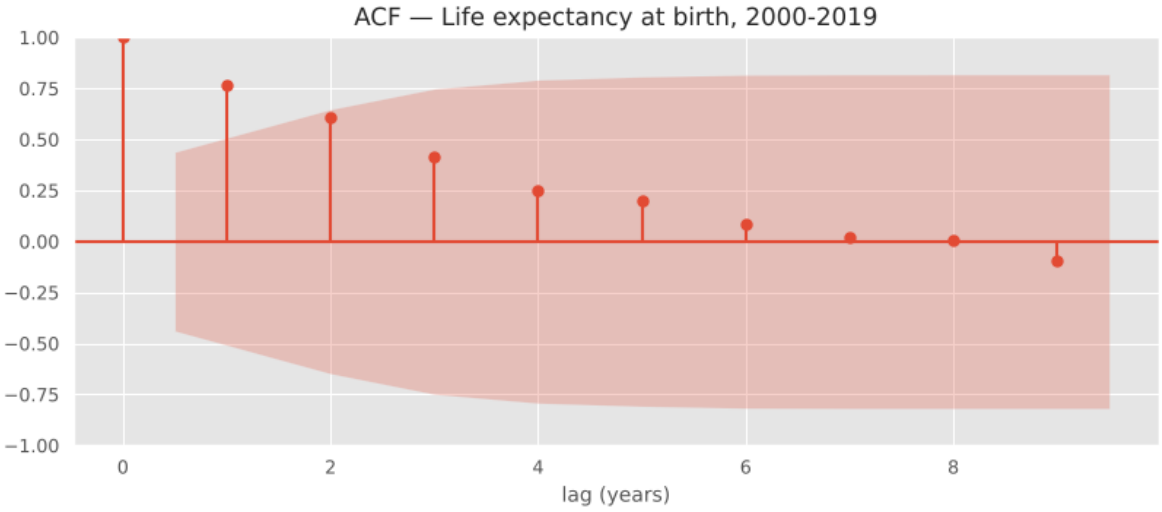


Figure 7. ACF of life expectancy, 2000–2019.

At lag 1 the points follow the diagonal; from lag 4 onward they move above it, meaning that life expectancy was usually higher than four or more years earlier. That is the upward trend, not a cycle:

there is no lag at which the points return to the diagonal, which is what a repeating pattern would produce. The ACF agrees: it is high at lag 1 (0.77) and decreases gradually (0.61, 0.41, 0.25...), the pattern of a series with persistence and a trend.

3. Top 10 causes of death (2000–2023)

The causes of death show a clear shift toward chronic disease: ischaemic heart disease and diabetes have been the two leading causes for most of the period, and both of their death rates more than doubled in 24 years. Fourteen different causes appear in the yearly top 10; nine of them appear every year.

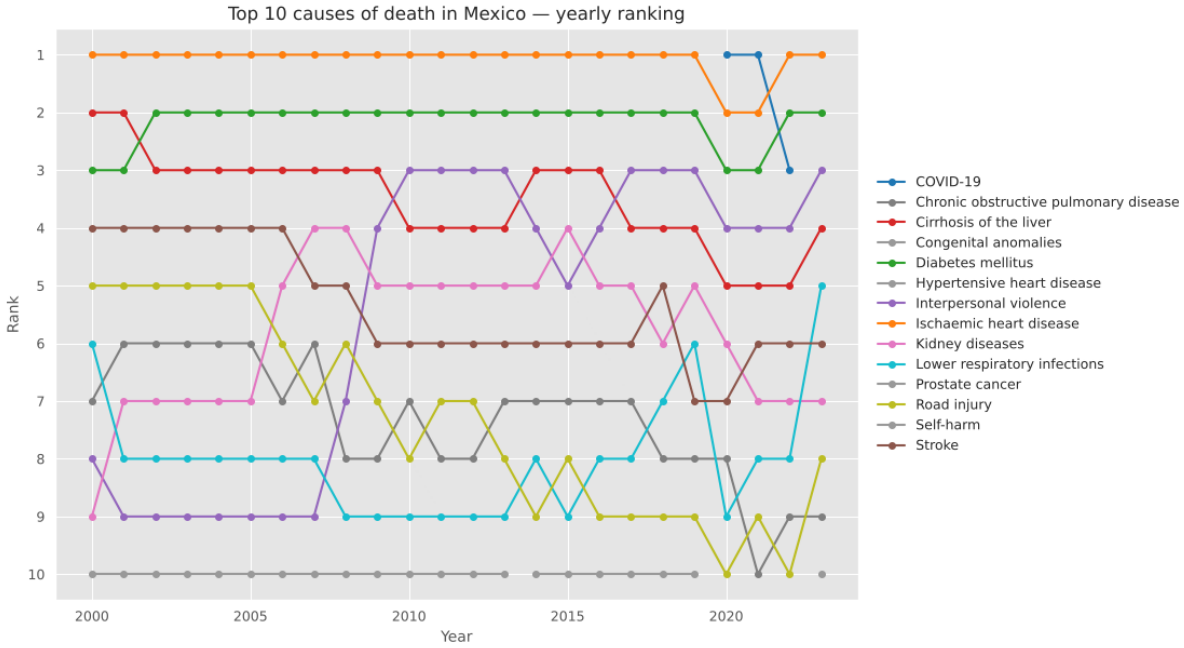


Figure 8. Ranking of the top 10 causes of death each year (1 = leading cause).

The ranking is remarkably stable at the top. Ischaemic heart disease is first every year except 2020 and 2021, and diabetes is second from 2002 onward (third in 2020–2021, when COVID-19 entered the list). Most movement happens lower down: interpersonal violence climbs from 9th to 3rd place in 2007–2010, lower respiratory infections rise to 5th place in 2023, congenital anomalies leave the list after 2013, hypertensive heart disease appears from 2014 to 2019 and self-harm enters in 2023.

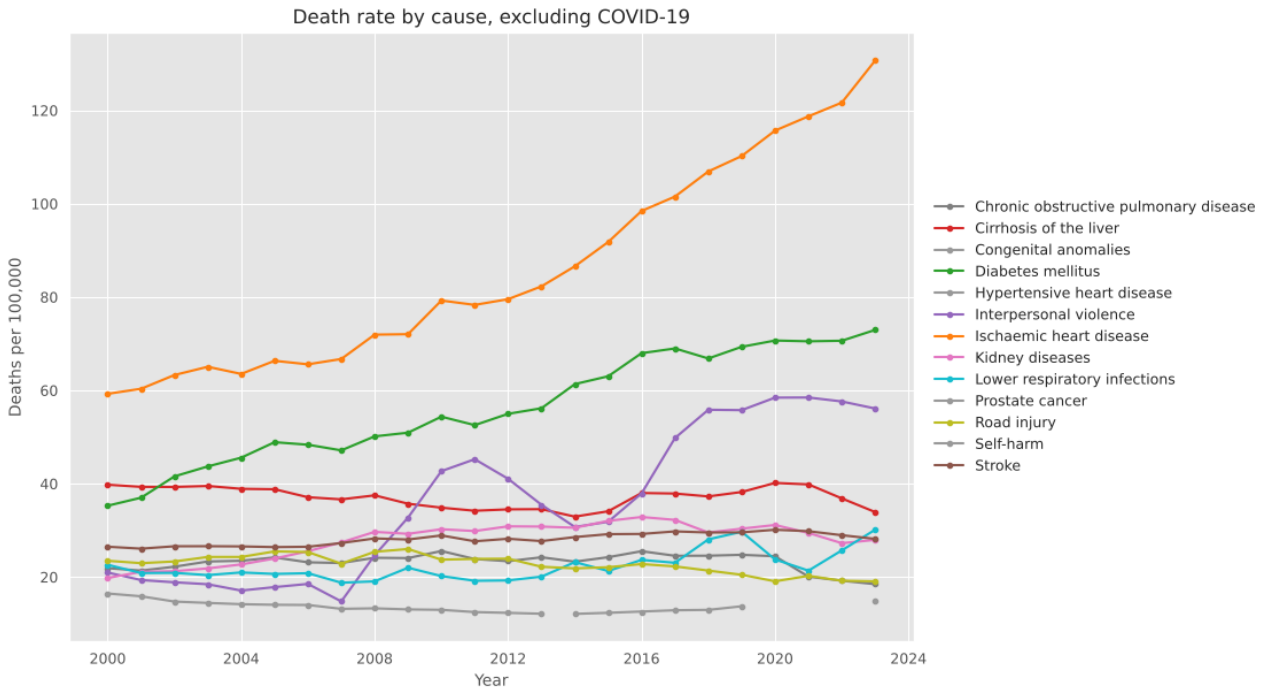


Figure 9. Death rates per 100,000 by cause, without COVID-19 so the other causes remain readable.

- **Ischaemic heart disease:** a strong upward **trend**, from 59.3 to 130.7 deaths per 100,000 (+120%). The increase speeds up after 2013 and continued during the pandemic (118.7 in 2021, 121.7 in 2022).
- **Diabetes:** a steady upward **trend**, from 35.3 to 73.0 (+107%), with a flat period around 70.6 between 2020 and 2022.
- **Interpersonal violence:** the most irregular series. It falls to its lowest value in 2007 (14.8), quadruples to 58.5 by 2021, with a temporary drop to about 31 in 2014–2015. These rises and falls do not repeat at a fixed interval, so they are event-driven episodes rather than a **cycle**.
- **Lower respiratory infections:** fall during the pandemic (29.7 in 2019, 21.4 in 2021) and rebound to 30.2 in 2023. A possible reason is that part of these deaths were recorded as COVID-19 in 2020–2021; the data alone cannot confirm it.
- **Stable causes:** cirrhosis, stroke, kidney disease, road injuries and COPD stay within narrow ranges for the whole period.
- **COVID-19 (outlier):** the leading cause in 2020 (345.3) and 2021 (344.8), almost three times the rate of heart disease; 70.0 in 2022 and out of the top 10 in 2023. It is the largest outlier in the dataset and lasted two full years.

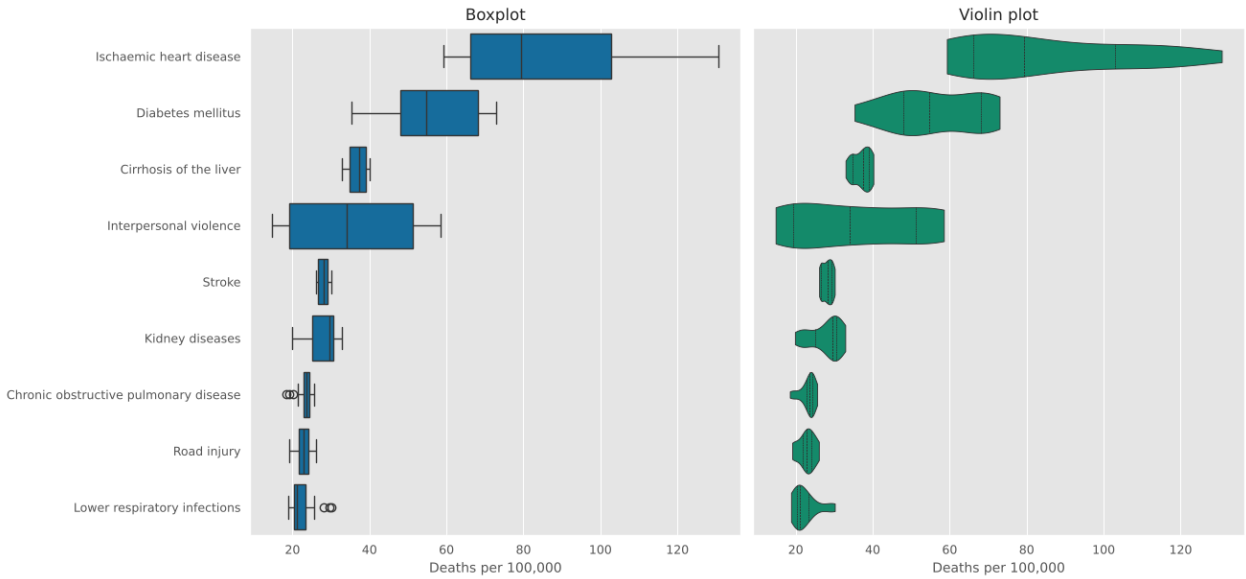


Figure 10. Distribution of each cause's yearly death rate (boxplot and violin plot), for the nine causes present in all 24 years.

The box shows where the middle 50% of the years fall, and its width (the interquartile range) shows how much a cause changed over time. Heart disease, diabetes and violence have wide boxes because they changed a lot; cirrhosis, stroke, road injuries and COPD have narrow boxes because they barely moved. The violin plots add the density: violence has two concentrations of values (about 15–21 before 2008 and about 50–58 from 2017 onward), which confirms that it shifted between two levels rather than changing gradually. The few points outside the whiskers are recent years: the low COPD values of 2022–2023 and the high respiratory-infection values of 2019 and 2023.

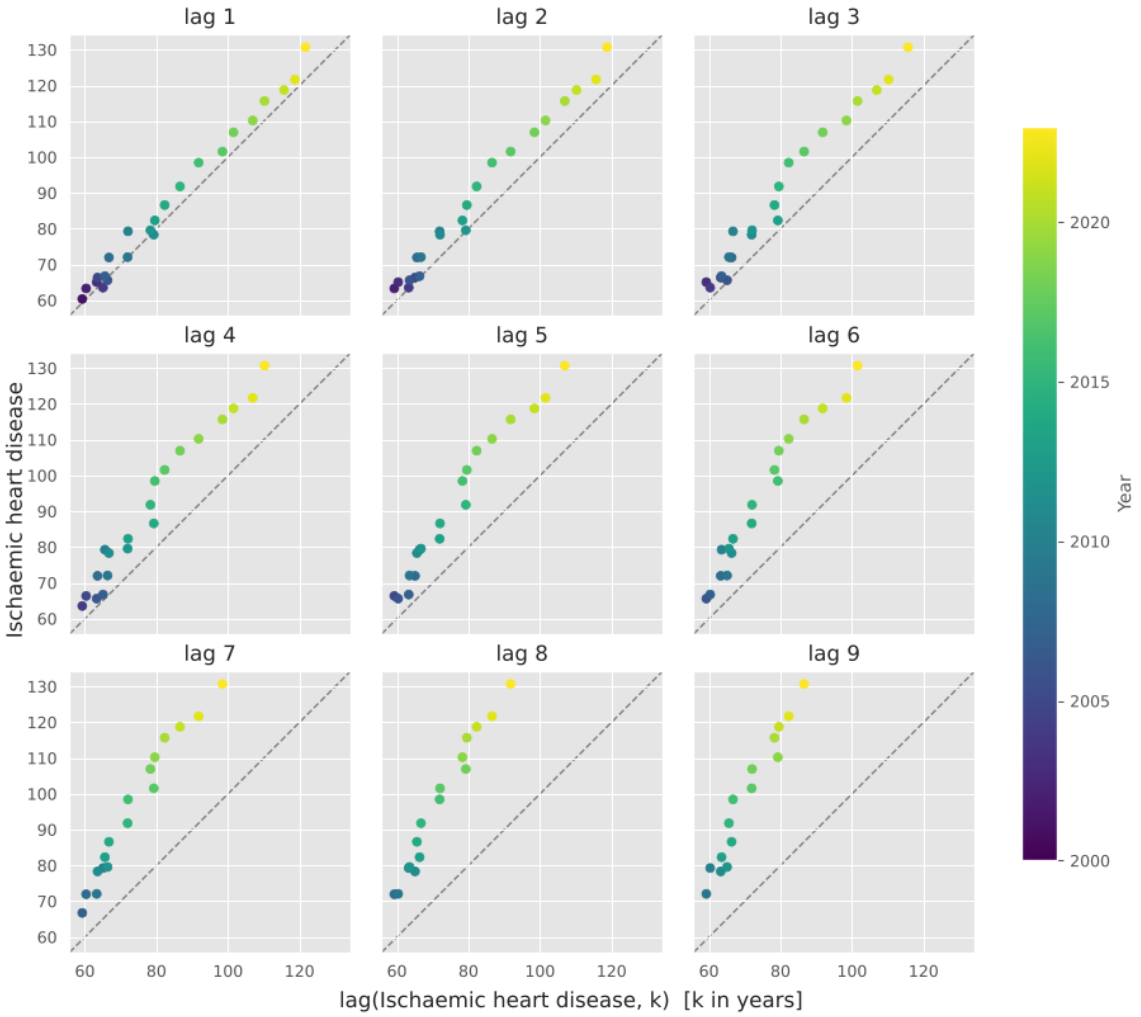


Figure 11. Lag plots of the death rate from ischaemic heart disease.

The points form a straight line that moves further above the diagonal as the lag grows: the death rate is always higher than it was k years earlier. This is the signature of a persistent upward trend. Diabetes shows the same pattern (see the notebook).

One limitation applies to this dataset: it contains only each year's top 10, so a cause that drops out of the ranking leaves gaps. These gaps are not missing data, but they mean that only the nine causes present every year form complete time series.

4. Risk factors

The risk factors linked to the leading causes of death are not moving together: adult obesity and alcohol consumption are rising, while tobacco use and hypertension are falling, and access to safe drinking water improves only slowly.

All five indicators are WHO modelled estimates, not raw survey results. That is why the curves are so smooth and the remainders are almost zero; the smoothness comes from the estimation method, and the shaded uncertainty intervals should be read together with each line.

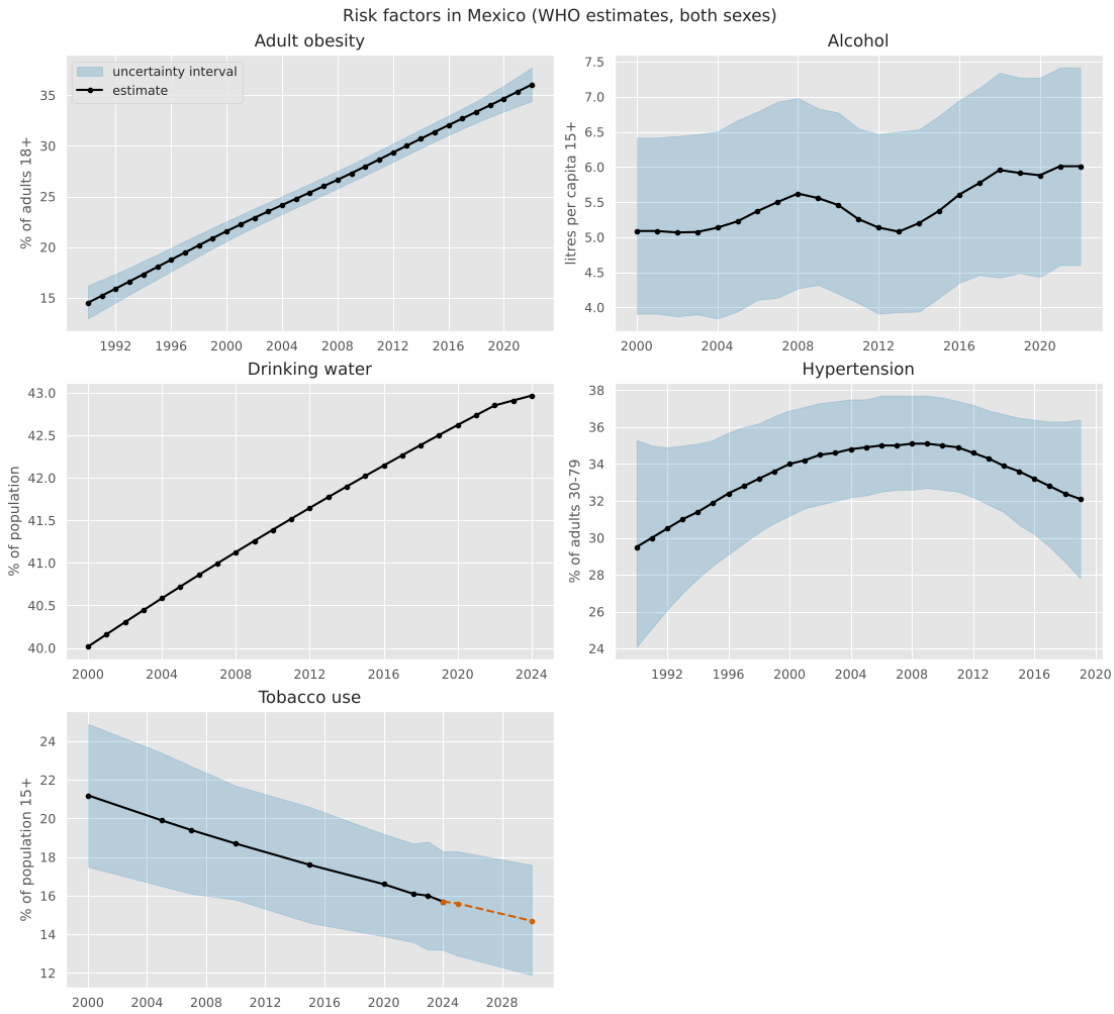


Figure 12. The five risk factors, both sexes, with WHO uncertainty intervals where available. Dashed orange = WHO projections (tobacco, 2025 and 2030).

Risk factor	First value	Latest value	Change	Shape of the series
Adult obesity, 18+ (% of adults)	14.5 (1990)	36.0 (2022)	+21.5 points	Almost perfectly linear rise, about +0.65 points per year
Hypertension, 30–79 (% of adults)	29.5 (1990)	32.1 (2019)	+2.6 points	Rise to a peak of 35.1 in 2008–2009, then decline
Alcohol, 15+ (litres per capita)	5.1 (2000)	6.0 (2022)	+0.9 litres	One wave: up to 2008, down to 2013, up again
Tobacco use, 15+ (% of population)	21.2 (2000)	15.7 (2024)	−5.5 points	Steady decline, irregularly spaced observations
Safely managed drinking water (% of population)	40.0 (2000)	43.0 (2024)	+3.0 points	Slow, almost linear rise

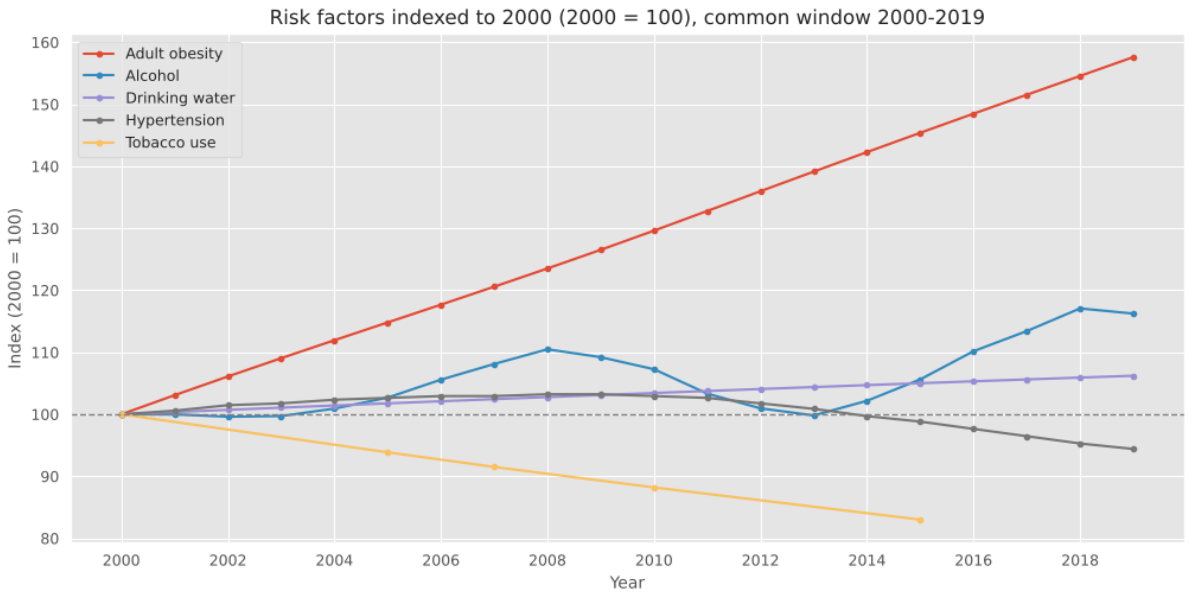


Figure 13. Risk factors indexed to their 2000 value (2000 = 100) over the common window 2000–2019, so that indicators with different units can be compared.

Indexing puts all five on the same scale. Obesity stands apart: it grew 58% between 2000 and 2019, far faster than any other factor. Alcohol rose 16%, drinking water 6%, while hypertension fell 6% and tobacco use 17% (to 2015, its last observation inside this window).

- **Trend:** every series is dominated by a trend. Obesity and drinking water rise in almost straight lines; tobacco falls in an almost straight line.
- **Change of pattern:** hypertension is the only factor whose direction reverses, from rising to falling around 2009.
- **Seasonality:** not applicable with annual data.
- **Cycles:** none. Alcohol is the only series with an up-down-up shape, but it is a single wave over 20 years, too short to call a cycle.
- **Outliers:** none. The modelled estimates do not show the pandemic years as unusual.
- **Data quality:** alcohol repeats the previous year's value in 2001 and 2022 (most likely carried forward), and tobacco includes projections for 2025 and 2030, which were excluded from the statistics.

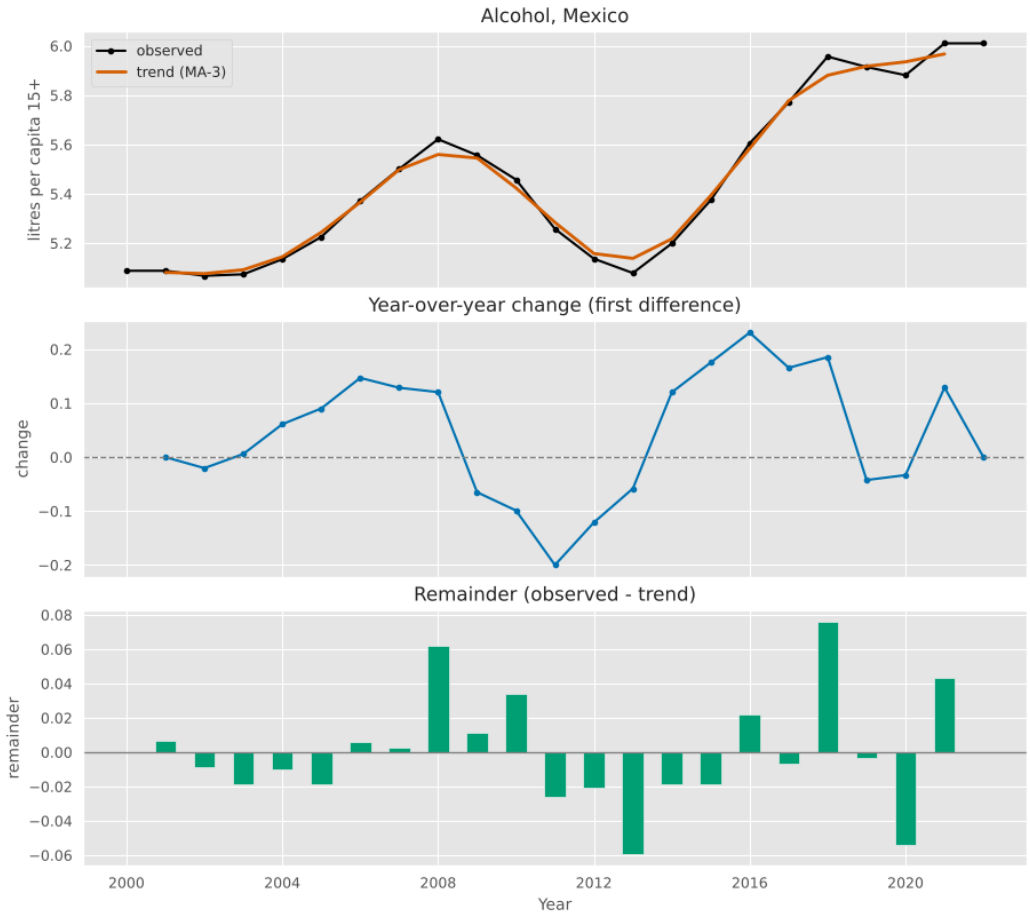


Figure 14. Alcohol consumption per capita (15+): trend, year-over-year change and remainder.

Alcohol is the most interesting of the five. Consumption rose from 5.1 to 5.6 litres between 2003 and 2008, fell back to 5.1 by 2013 and then climbed to 6.0 litres. The year-over-year changes are small (at most ±0.2 litres), well inside the uncertainty interval of roughly ±1.3 litres, so the wave should be read as a tendency rather than a precise measurement.

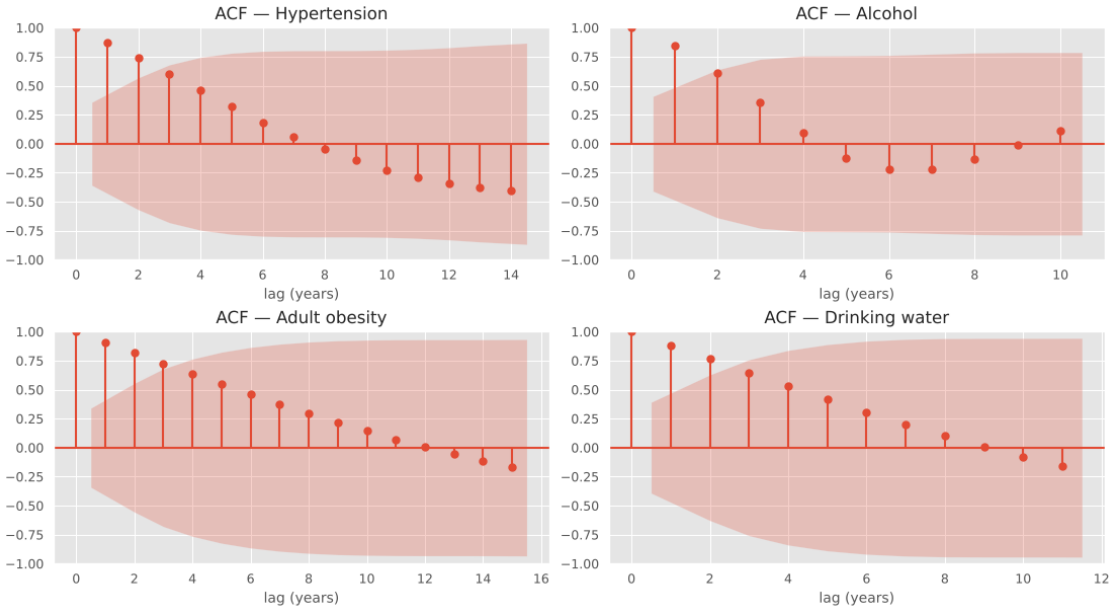


Figure 15. ACF of the four annual risk factors (tobacco is excluded because its observations are irregular).

Obesity, drinking water and hypertension show slowly decaying ACFs, the signature of a trend. Hypertension turns negative at long lags because the series rose and then fell. Alcohol drops faster and becomes negative around lags 6–7, reflecting its single wave. Lag plots for these series (in the notebook) tell the same story: points lie along the diagonal at lag 1 and drift away from it at longer lags.

Conclusions

The data describe a country in the middle of a health transition: Mexico's population is growing more slowly and living longer than in the past, but its gains in life expectancy stalled before the pandemic, and mortality is increasingly driven by chronic diseases.

1. **Trend is the dominant feature of every series.** Each year is very close to the previous one (strong lag-1 relationships and slowly decaying ACFs), so all the series are non-stationary. Population growth falls steadily from its 1966 peak; heart disease, diabetes and obesity rise steadily.
2. **There is no seasonality and no regular cycle.** Seasonality cannot be observed in annual data, and the irregular episodes that do appear (violence in 2008–2011 and after 2015, the alcohol wave, the reversal of hypertension) do not repeat at a fixed interval. With 20 to 74 observations per series, any cyclic claim would not be supported by the data.
3. **COVID-19 is the main outlier and structural break.** It was the leading cause of death in 2020 and 2021 at about 345 deaths per 100,000, and it cut life expectancy by 4.9 years in one year. The shock lasted two full years in the data.
4. **The pandemic interrupted life expectancy, not the rise of chronic disease.** Heart disease deaths kept climbing through 2020–2023 and diabetes deaths stayed at their highest level. The long-term problem existed before COVID-19 and continues after it.
5. **Risk factors are consistent with the mortality profile.** The steady rise of adult obesity (from 21.6% to 34.0% between 2000 and 2019) runs alongside the doubling of heart disease and diabetes deaths. This is an association between trends, not proof of cause and effect, but it shows where prevention could matter most.
6. **Some indicators improve.** Tobacco use has declined for two decades, hypertension prevalence has fallen since 2009 and access to safe drinking water is rising, although 57% of the population still lacks safely managed water.
7. **The data are not up to date.** Life expectancy ends in 2021, hypertension in 2019 and causes of death in 2023. Decisions taken today rely on figures that are two to seven years old, and the post-pandemic recovery is not yet visible in the WHO life expectancy series.

For decision-makers, the message is that the pandemic was a temporary shock, while chronic disease linked to obesity and lifestyle is a long-term trend that is still growing.

Appendix

A. Datasets used

All files were downloaded from the WHO country page for Mexico (data.who.int/countries/484); only the total (both sexes) values were used. The script reads them from the same folder.

File	Content	Years	Rows used
Population mexico.xlsx	UN World Population Prospects estimates for Mexico; column "Population Growth Rate (percentage)"	1950–2023	74
Life expectancy at birth.xlsx	WHO WHOSIS_000001, life expectancy at birth (years)	2000–2021	22 (TOTAL)
causas_muerte_mexico_2000_2023_OMS.csv	Top 10 causes of death per year, deaths per 100,000	2000–2023	240
Prevalencia de hipertencion en adultos entre 30 y 79.xlsx	WHO NCD_HYP_PREVALENCE_A (% of adults 30–79)	1990–2019	30 (TOTAL)
Prevalencia de uso de tabaco en mayores de 15.xlsx	WHO M_Est_tob_curr_std (% of population 15+)	2000–2030	11 (TOTAL)
Consumo de alcohol mayores de 15.xlsx	WHO SA_0000001688 (litres of pure alcohol per capita, 15+)	2000–2022	23
Obesidad en adultos mayores.xlsx	WHO NCD_BMI_30A (% of adults 18+)	1990–2022	33 (TOTAL)
población que utiliza servicios de agua potable gestionados de forma segura.xlsx	WHO/UNICEF JMP WSH_WATER_SAFELY_MANAGED (% of population)	2000–2024	25

The data files are attached with this report.

B. Python script

The complete analysis is in the Jupyter notebook **EDA_WHO_Mexico.ipynb**, attached with this report. Placed in the same folder as the data files, it reproduces every table and figure shown here when run from top to bottom (Run All).

Notebook section	Content
1–3	Data loading, coverage, data-quality checks, summary statistics
4	Helper functions: trend + remainder plot, lag plots, ACF
5	Population growth rate
6	Life expectancy at birth
7	Top causes of death

Notebook section	Content
8	Risk factors

Libraries: pandas, numpy, matplotlib, seaborn, statsmodels (ACF) and openpyxl (to read the Excel files).